PS-6 run report: calibration, five timed ecbench sessions, pair-table memory

`research/pdp_oracle_speedup_20261011/ps6_charge/raw/` — 2026-10-31 from 19:24 UTC (calibration, then the timed sessions to 20:16; tests, replays and memory runs after)

1. What ran

`ps6_charge/run_ps6.sh` stages `calibrate`, `timed`, `reaudit`, `tests` and `memory`, inside one hold of the shared benchmark lock for the timed work (`tools/isolated_bench.py busy`), with ecbench's own whole-core reservation (`ecbench run --cpus auto`, a private inner lock). Binary SHA-256 `c8a7e9b9000f...`, commit `7ded1ca84` (clean tree). Host: Intel Xeon 2.10 GHz KVM guest, 4 vCPU, kernel 6.18.44-fc-v185 (`raw/host.txt`, `raw/manifest.json`). Specs `ps6_charge/specs/timed_n15 .. n27.json`: arms `rho-negation`, `s4-baseline`, `s4-baseline-control` (the baseline repeated, the A/A), `s4-grouplaw`, `mitm`, `mitm-negfold`; 8 targets x 5 rounds + 1 warm-up (4 targets at $n = 27$); alternate order.

Runs requested and completed. 288 + 288 + 288 + 288 + 144 = 1,296 requested, 1,296 completed and verified (1,080 measured, 216 warm-up), 0 failed, 0 timed out, 0 refused starts. Audits: all `ok` (`--replay-all` at $n \leq 21$: 240 replays each; `--replay 12` at $n = 24, 27$). Calibration (`tools/isolated_bench.py run --cpus 3`, 58 s, not contended): 11 interleaved repetitions per rung. Unit tests and ecbench tests (`raw/logs/tests*.log`): `--bin ecbench ps6` 4 passed; `--lib ecbench` 102 passed; `--test ecbench` 16 passed.

Isolation levels earned (per run, measured rows): $n = 15$: 220 at L2, 68 at L1; 18: 217, 71; 21: 209, 79; 24: 181, 107; 27: 64, 80. The only L2 blockers recorded were one or more hypervisor steal ticks on the run CPU (389 runs) and run-queue delay above 0.5% of the solve (18 runs); L3 needs a lab host.

2. Observations

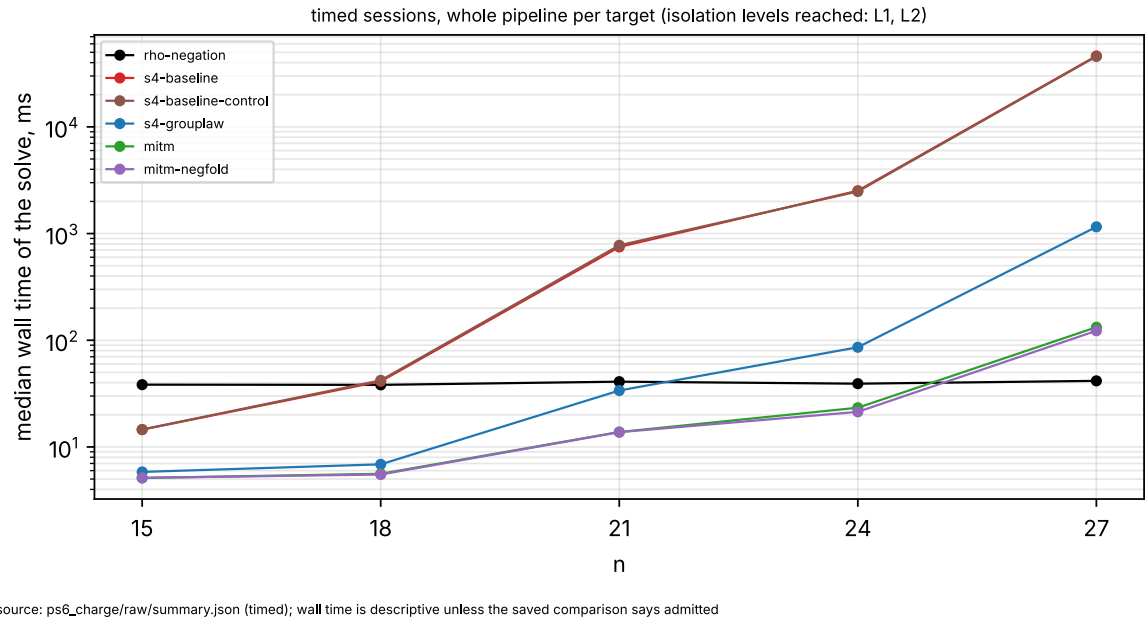


Figure 1: Median wall time of the solve per run, by arm and rung, from the timed sessions (source `ps6_charge/raw/summary.json`, `timed`; isolation levels L1 and L2).

n	baseline ms	kernel ms	MITM ms	rho ms	A/A: control / baseline
15	14.5	5.8	5.1	38.4	0.999 (0.966–1.037)
18	41.2	6.9	5.6	38.2	1.014 (0.984–1.056)
21	749	33.8	13.8	40.9	1.019 (0.993–1.055)
24	2,521	86.0	23.3	39.2	1.006 (0.983–1.017)
27	46,151	1,156	132.5	41.6	0.999 (0.991–1.010)

Medians over 40 runs per arm (20 at $n = 27$); the A/A column is the paired median of the baseline repeated over the baseline, with a 95% cluster-bootstrap interval over targets. Paired ratios with intervals for every arm pair, the isolation levels of the pairs and ecbench's own `admitted` / `descriptive` status are in `raw/summary.md` and the saved comparisons under each session's `comparisons/`. Kernel over baseline, paired median (95% interval): 0.385 (0.363–0.444), 0.163 (0.154–0.179), 0.044 (0.043–0.046), 0.035 (0.034–0.036), 0.025 (0.024–0.025) at $n = 15..27$. At $n = 27$ every pair that includes the baseline is L1 (no

L2 pair), so ecbench calls those comparisons descriptive; the same ratios at $n \leq 21$ include L2 pairs. Rho's wall is 38–42 ms at every n.

Memory. `ps6_tool memory` built the library's pair table on each rung's base: 220, 1,016, 3,721, 18,711 and 69,118 entries at $n = 15..27$; resident growth 0.0, 0.0, 0.1, 0.5 and 2.1 MiB (full and negation-folded builds give the same entry counts and 465 to 138,601 vs 240 to 69,432 group additions); a run's peak resident set is 100–108 MiB in every arm (`max_rss_kib` in each record).

Replays. The post-merge binary replayed the five counted sessions of the earlier run (all reproduced).

3. Not in this report

No end-to-end claim and no transfer to any larger rung or curve. A baseline run takes 46 s at $n = 27$, so a rung above 27 was not attempted in this run.